Track: Formula 1 Track

Tools:

- Hardware:
  - Arduino Uno
  - MFRC522 RFID Array
  - WebCam
  - BreadBoard
  - Jumper Wire Assortment pack
- Software:
  - Python
  - MySQL
  - Unity
- Physical tools:
  - Race Track (miniature)
  - 8 Cars (miniature)
- Libraries:
  - OpenCV
  - OpenCV Contrib
  - PySerial

In our project we will build a custom hybrid sensor pipeline using a miniature physical environment using 8 toy cars and a custom track layout as a live dataset generator. The software will use physical motion data in 2 systems:

- The Dashboard: A broadcast style hub featuring responsive speedometers, live sector tracking, and graph analytics.
- Racing mode: Users will step directly onto a digital track to actively race their own virtual car against the live tracked physical cars, with the option to race in old Formula 1 races.

To implement the systems we have designed a 3 layers platform:

- The physical layer (IoT)
  - Webcam tracking: we will have a webcam that will stream the visual matrices at 60FPS to calculate the continuous coordinates of the cars.
  - ArUco Binary Markers: Printed square matrices fixed onto the roofs of the 8 cars, utilized by the software to determine unique orientation and ID vectors without adding physical weight.
  - Arduino Checkpoints: We will use Arduino Uno R3 wired using a shared SPI bus to an MFRC522 RFID Sensor Array hidden underneath the track. Every toy car will have a paper thin RFID tag embedded with a hardcoded unique identifier (UID).
- The Middleware layer
  - Python: Running concurrent data, gathering protocols.
  - OpenCV: Process the webcam frames and read the ArUco markers to calculate the velocity.
  - PySerial: Listens to the Arduino's USB COM.
  - The Algorithm: The webcam will provide a high frequency positioning data, but its prone to visual noise. So we have the physical RFID gate, when a car crosses above it, the Arduino will transmits its ID and timestamp. Python will infer the data and will instantly override the webcam's estimated position.
- The Frontend layer
  - Unity: Reads the data packets parsed from python to dynamically alter the coordinate matrices of 3D vehicle assets frame by frame.

This three layer architecture is fully scalable to a real Formula 1 environment. We only need to swap the webcam and RFID checkpoints with the tools that are used in Formula 1 tracks.